Week 8 Tutorial Solutions

Problem 34

Consider the finite region bounded by the functions $f(x) = \dfrac{x}{3}$ and $g(x) = \sqrt{x}$.

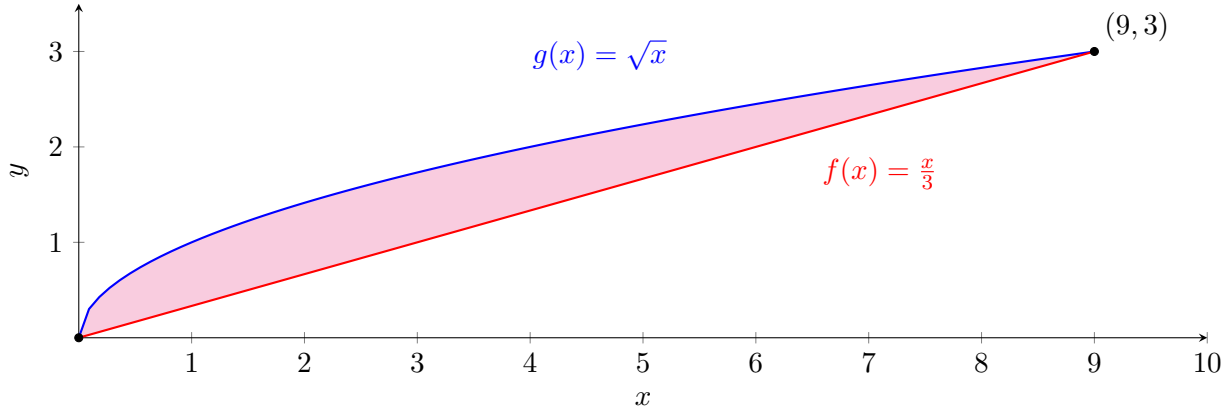
(a) Sketch the graphs of $f(x)$ and $g(x)$, and then shade in the region defined by those boundaries. Clearly indicate the scale on both axes.

(b) Write an integral that represents the area enclosed between these two graphs, computed using **vertical** slices, Δx or dx thick. You do **not** need to evaluate this integral.

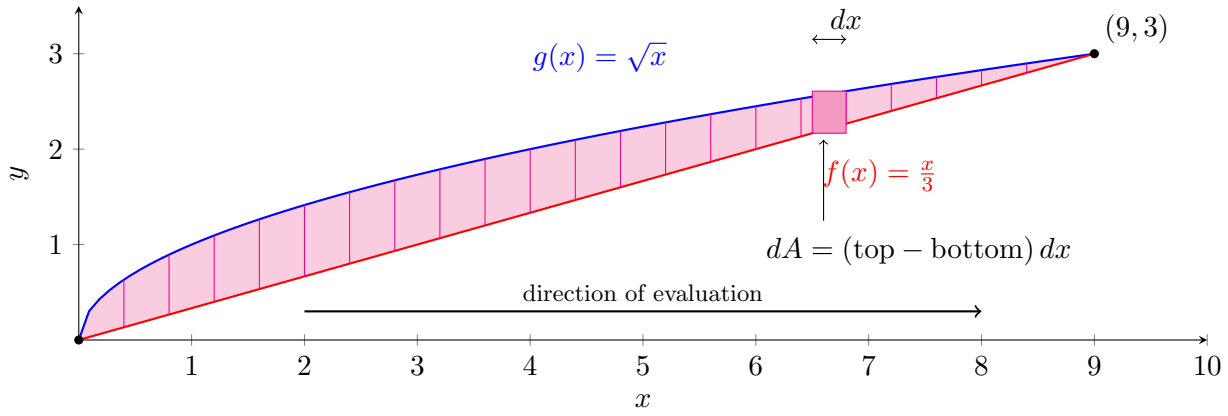
(c) Write an integral that represents the area enclosed between these two graphs, computed using **horizontal** slices, Δy or dy thick. You do **not** need to evaluate this integral.

Solution

(a)



(b) Vertical slices: integrate top − bottom, moving left to right.



Find the points of intersection:

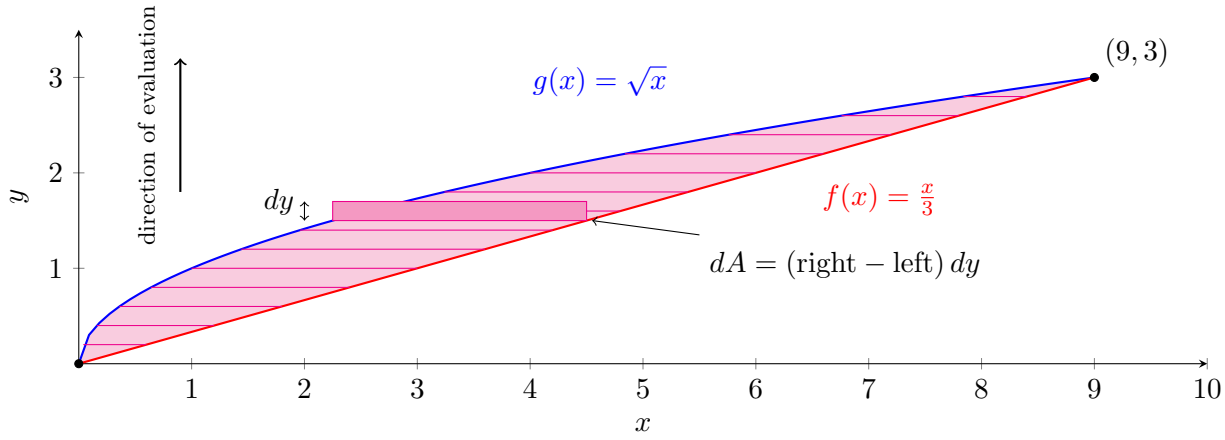
$$\frac{x}{3} = \sqrt{x} \quad \Longrightarrow \quad \frac{x^2}{9} = x \quad \Longrightarrow \quad 0 = 9x - x^2 = x(9-x) \quad \Longrightarrow \quad x = 0 \text{ or } x = 9.$$

The slices have thickness dx and run from $x = 0$ to $x = 9$, with $g(x)$ on top and $f(x)$ on the bottom:

$$A = \int (\text{top} - \text{bottom})\,dx = \int_0^9 \big(g(x) - f(x)\big)\,dx$$

$$\boxed{A = \int_0^9 \left(\sqrt{x} - \frac{x}{3}\right) dx}$$

(c) Horizontal slices: integrate right − left, moving bottom to top.



The slices run from $y = 0$ to $y = f(9) = \dfrac{9}{3} = 3$. Writing each function in terms of y:

$$y = \frac{x}{3} \implies x = 3y, \qquad y = \sqrt{x} \implies x = y^2.$$

The right boundary is $x = 3y$ and the left boundary is $x = y^2$:

$$A = \int (\text{right} - \text{left})\, dy$$

$$\boxed{A = \int_0^3 \left(3y - y^2\right) dy}$$

Problem 35

Evaluate the indefinite integral $\int xe^{-4x}\,dx$. Determine by differentiating whether or not your answer is correct.

Solution

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Let

$$u = x \implies du = dx, \qquad dv = e^{-4x}\,dx \implies v = -\frac{1}{4}e^{-4x}.$$

Then

$$\begin{aligned}\int xe^{-4x}\,dx &= (x)\left(-\frac{1}{4}e^{-4x}\right) - \int -\frac{1}{4}e^{-4x}\,dx \\ &= -\frac{x}{4}e^{-4x} - \left(-\frac{1}{4}\right)\left(-\frac{1}{4}\right)e^{-4x} + C\end{aligned}$$

$$\boxed{\int xe^{-4x}\,dx = -\frac{x}{4}e^{-4x} - \frac{1}{16}e^{-4x} + C}$$

Check by differentiating. Let $F(x) = -\frac{x}{4}e^{-4x} - \frac{1}{16}e^{-4x}$. Then

$$\begin{aligned}F'(x) &= -\frac{1}{4}e^{-4x} - \frac{x}{4}\left(-4e^{-4x}\right) - \frac{1}{16}\left(-4e^{-4x}\right) \\ &= -\frac{1}{4}e^{-4x} + xe^{-4x} + \frac{1}{4}e^{-4x} \\ &= xe^{-4x}.\end{aligned}$$

Since $F'(x) = xe^{-4x}$, the answer is correct.

Problem 36

Evaluate the indefinite integral $\int \frac{2+x}{1+x^2}\,dx$. Determine by differentiating whether or not your answer is correct.

Solution

Split the integrand:

$$\int \frac{2+x}{1+x^2}\,dx = \int \frac{2}{1+x^2} + \frac{x}{1+x^2}\,dx = 2\int \frac{1}{1+x^2}\,dx + \int \frac{x}{1+x^2}\,dx.$$

The first integral is the antiderivative of $\arctan(x)$'s derivative:

$$2\int \frac{1}{1+x^2}\,dx = 2\arctan(x) + C_1.$$

For the second integral, use the substitution $u = 1+x^2$, so $\dfrac{du}{dx} = 2x$:

$$\int \frac{x}{1+x^2}\,dx = \int \frac{x}{u}\frac{du}{2x} = \frac{1}{2}\int \frac{1}{u}\,du = \frac{1}{2}\ln|u| + C_2.$$

Therefore

$$\boxed{\int \frac{2+x}{1+x^2}\,dx = 2\arctan(x) + \frac{1}{2}\ln\left|1+x^2\right| + C}$$

Check by differentiating.

$$\frac{d}{dx}\left[2\arctan(x) + \frac{1}{2}\ln\left|1+x^2\right|\right] = 2\left(\frac{1}{1+x^2}\right) + \frac{1}{2}\left(\frac{1}{1+x^2}\right)(2x) = \frac{2+x}{1+x^2}.$$

This matches the original integrand, so the answer is correct.

Problem 37

Evaluate the following integrals:

(a) $\displaystyle\int xe^{-3x^2}\,dx$

(b) $\displaystyle\int \arccos(x)\,dx$

Solution

(a) Use the substitution $u = -3x^2$, so $\dfrac{du}{dx} = -6x$:

$$\int xe^{-3x^2}\,dx = \int xe^{u}\,\frac{du}{-6x} = -\frac{1}{6}\int e^{u}\,du = -\frac{1}{6}e^{u} + C.$$

$$\boxed{\int xe^{-3x^2}\,dx = -\frac{1}{6}e^{-3x^2} + C}$$

(b) Use integration by parts, $\displaystyle\int u\,dv = uv - \int v\,du$.

Let

$$u = \arccos(x) \implies du = \frac{-1}{\sqrt{1-x^2}}\,dx, \qquad dv = dx \implies v = x.$$

Then

$$\int \arccos(x)\,dx = x\arccos(x) - \int \frac{-1}{\sqrt{1-x^2}}\,(x)\,dx.$$

For the remaining integral, use the substitution $w = 1 - x^2$, so $\dfrac{dw}{dx} = -2x$:

$$\int \frac{-x}{\sqrt{1-x^2}}\,dx = \int \frac{-x}{\sqrt{w}}\,\frac{dw}{-2x} = \frac{1}{2}\int w^{-1/2}\,dw.$$

Therefore

$$\int \arccos(x)\,dx = x\arccos(x) - \frac{1}{2}\int w^{-1/2}\,dw = x\arccos(x) - \frac{1}{2}\cdot\frac{w^{1/2}}{1/2} + C$$

$$\boxed{\int \arccos(x)\,dx = x\arccos(x) - \sqrt{1-x^2} + C}$$